Towards Rigorous Role Design: A Survey Study on the Role Perceptions of Digital Twins in Preventive Health Contexts

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Abstract

As Human Digital Twins (HDTs) evolve into lifelong health companions, role design has appeared as a strategy to structure multifaceted interactions. However, empirical evidence of its impact on user perception is scarce. Through an online survey (N=263), we explored perceptions of 12 HDT roles within 3 groups: Self, Prediction Maker, and Data Explorer. Participants evaluated each role with preventive health scenarios across 5 perceptual dimensions and completed 3 intra-group rankings. Our initial results suggest a notable tension between the central "twinning" metaphor and the significantly lower scores of the Self group across four dimensions. Notably, Self-related roles did not elicit higher perceived emotional support than others, and high perceptual scores did not consistently translate into high rankings. These patterns suggest that emotional support may be considered as a tradeoff dimension rather than a primary expectation, inviting discussions on rigorous role selection that goes beyond data mimicry in health technology design.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

Survey, Role Design, Perceptions, Digital Twin, Preventive Health, Use Study, User Experience Design

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1 Introduction

Human Digital Twins (HDTs), defined as digital replicas of individuals in the physical-world [15], are increasingly positioned as next generation health technologies with advanced simulation and predictive capabilities offering insights in medical [13], rehabilitation [14] and everyday preventive health contexts [21]. Unlike conventional self-tracking or health monitoring tools, HDTs have been envisioned as "future data selves" [7], "sidekicks" [5] or "digital doppelgängers" [11] that continuously evolve with the user, deeply entangling with personal identity, relationships with technologies, and personal health practices. To structure this complexity in interaction design, role design, the use of specific metaphors to structure the technology's capability and relationship with users [18], has appeared as a useful strategy.

While "assistant", "companion" or "friend" are widely adopted in social robotics [20], artificial intelligence [12], and chatbots [23], role framings in the HDT field remain largely conceptual, ranging from "facilitators of longevity" to "24/7 assistants" [5, 6]. This conceptual ambiguity obscures what is distinctive about role design for HDTs, whose metaphors often imply a self-relation involving digital self-representation, longitudinal health prediction and user agency. Thus, we argue that how HDTs are framed through roles become central to how users interpret, trust, and engage with these emerging digital counterparts. However, it remains unclear how these framings influence user perception or whether users can differentiate among designed metaphors (e.g., a "Twin" vs. a "Shadow"). To address this, we adopt a user-centric approach to investigate the following research questions:

RQ1: To what extent do users differentiate among HDT roles and role groups across five perceived dimensions (comfort, emotional support, trust in giving prediction, advice-following, and desirability)?

RQ2: How do these ratings influence users' role preferences in preventive health contexts?

We conducted a within-subject online survey (N=263) in which participants evaluated their perception of 12 scenario-based roles on 5 perceptual dimensions. Rather than seeking a single "ideal" role, this paper offers an initial empirical account of role framing in HDT design for preventive health purposes. It explores how different role framings may influence people's perceptions, from perceived comfort to perceived emotional support, thereby opening up discussions on rigorous role design of HDTs and adjacent health technologies.

2 Related Work

2.1 Human Digital Twins in Healthcare

Existing research on HDTs in healthcare has largely focused on its technical feasibility and patient health modeling in clinical settings [2, 14], leaving the layer of interaction design underexplored. While some early explorations of HDTs in health contexts, such as Shengli's [22] life-course scenario, outlined a system structure

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for data integration and pointed to potential complexities and societal impacts, they did not address how interaction design might mitigate ethical tensions or support sustained engagement. Emerging design-oriented studies have begun to bridge this gap. For instance, Lauer-Schmaltz et al. [14] identified trust and meaningful embodiment as essential design themes for fostering human–HDT relationships. However, these high-level principles remain difficult to translate into concrete interaction design strategies. Similarly, qualitative insights from De Maeyer et al. [5, 6] suggest that users envision HDTs as "a digital companion" or "sidekick", emphasizing that "life is a conversation, not a database", yet these framings have not been evaluated systematically. Consequently, there is limited quantitative evidence regarding which role framings align with user perceptions and preferences in practice.

2.2 Role Design in Digital Health Technology

Role framing has been widely used in digital health technologies to structure interaction design and meet users' health requirements. Prior studies in conversational agents [9, 19] and social robots [4, 20] often adopt roles such as coach, assistant, expert, or companion to facilitate engagement, trust, and perceived emotional support. A recent study [12] comparing interacting with two customized ChatGPT personas further suggests that for a successful quasi-social relationship, machines should authentically assume roles that "naturally" fit their non-human nature while ensuring the user remains in control.

Despite the prevalence of role metaphors, few studies have systematically outlined and compared different role framings and people's perceptions of them. In domains adjacent to HDTs, Ringfort-Felner et al. [20] categorized four role groups that move beyond "subservient assistant" in social robotics, and Agrawal et al. [1] identified four collaborative roles Digital Twins can perform in industrial settings. However, applying these role framings directly to HDTs is complicated by their potentially close connections with a specific user's biological identity, personal health data and behaviors. This connection requires role metaphors to be chosen carefully, as they may do more than merely facilitate interaction, but may also reflect and shape how users perceive themselves and their own health. Understanding how users perceive different role framings is therefore essential for designing appropriate HDT interactions.

3 Method

3.1 Survey Design

3.1.1 Roles and Scenarios. The survey examined 12 roles derived from a prior conceptual study, in which the roles were developed and organized into three groups to capture distinct ways an HDT may be perceived by the user and how it support reflections on their health data. The three groups include: *Self*, *Prediction Maker*, and *Data Explorer*.

Roles in the *Self* group frames HDTs as supporting self-reflection and self-recognition within their health data. The *Prediction Maker* group frames HDTs as making health forecasts and support proactive decision-making. The *Data Explorer* group frames HDTs as allowing users to experiment with and make sense of their health data through the technology's temporal and analytical affordances. In the survey, participants were presented with a role description paired with a scenario in a preventive health context. An example is given in Table 1.

3.1.2 Survey. The survey first collected demographic information (age, nationality, gender, employment status, technology/design background, comfort with digital technology, and familiarity with digital health and Digital Twin concepts), followed by questions about participants' self-tracking experience. Participants were then presented with a description of Digital Twins and Human Digital Twins including their potential in preventive health contexts, which also served as an initial attention check question.

Participants evaluated 12 roles in randomized order. Each HDT role was evaluated using 7-point Likert scales (strongly disagree – strongly agree) across five dimensions: *comfort* ("I would feel comfortable interacting with a digital twin that behaves in this way"), *emotional support* ("This digital twin would provide strong emotional support"), *trust in giving prediction* ("I would trust this digital twin to make health-related predictions for me"), *advice-following* ("I would be likely to follow this digital twin's health advice"), and *desirability* ("I would find it desirable to use this digital twin in daily life"). Finally, participants ranked the four roles within each group and answered two open-ended questions. The five dimensions were designed to provide a preliminary understanding of user perceptions rather than as standardized psychometric instruments, drawing inspiration from prior work on technology acceptance in the healthcare domain [10], factors of trust in health technology [17], user experience, and persuasive technology design [8].

3.2 Participants

We recruited 413 participants via the Prolific platform. To ensure data quality, we excluded incomplete responses, participants who failed any of the four attention check items, and responses submitted in less than half of the median completion time (Median = 16.6 minutes) [3]. We also screened responses with suspicious patterns using the LongString index [16]. It resulted in 263 valid answers. 5 participants left the open-ended question blank.

Participants (N = 263) ranged in age from 18 to 78 years (M = 42.73, SD = 13.125). Most were female (52.1%), 46.8% are male, and 1.1% are non-binary. They were mainly from Europe (55.9%), the USA (30.8%), Canada (5.3%) and Australia (3%). On a 5-point Likert scale from low to high, they have a high comfort level with digital technologies overall (M = 4.43, SD = 0.97) and moderate familiarity with digital health technologies (M = 3.80, SD = 0.93), but low familiarity with Digital Twins (M = 1.53, SD = 0.81).

3.3 Data Analysis

We conducted initial statistical analyses in SPSS. For the Likert-scale ratings, we first employed Linear Mixed-Effects Models (LMMs) to examine whether participants systematically differentiated among HDT roles. This approach was chosen to account for the repeated-measures study design, in which each participant evaluated multiple roles, resulting in non-independent observations. In the LMMs, *participants id* were specified as random intercepts to model individual differences in baseline rating tendencies, while *role* was treated as a fixed effect. Separate models were fitted for each perceptual dimension. Following the inferential analysis, descriptive statistics were computed to characterize the overall perceptual profiles, and

Table 1: Digital Twin Roles and Example Scenario.

Role Type	Description	Roles	Example Scenario (the Visionary Role)
Self	A reflective mirror of the user, simulating health data to enhance self-awareness.	Twin, Self Sculptor, Shadow, Persona	**Description:** It looks forward and helps you see long-term trends. It connects the dots between your current actions and future goals, showing how small changes today can lead to significant shifts in the future. **Scenario:** It shows you a bold future, a life far better than you imagined. "If you tweak just these small habits," it suggests, "look at what could unfold." It outlines a realistic yet inspiring vision of what's possible, helping you see potential outcomes in a motivating light. You feel drawn toward making those changes.
Prediction Maker	A forward-thinking guide that analyzes patterns to forecast future outcomes.	Pathfinder, Future Teller, Visionary, Forecaster	
Data Explorer	An analytical counterpart that organizes and supports experiments with data.	Librarian, Tester, Time Machine, Living Memory	

a group-level LMM was used to examine whether the corresponding role groups exhibited systematic perceptual differences across the Self, Prediction Maker, and Data Explorer groups. Repeated-measures ANOVA was not used because it assumes sphericity and homogeneity of variance across conditions, assumptions that are difficult to justify given the conceptual heterogeneity of 12 HDT roles. To enable interpretable comparisons across the large set of roles, post-hoc pairwise comparisons were conducted using estimated marginal means with Bonferroni-adjusted significance thresholds. These comparisons were applied to examine role-level differences following the group-level LMMs.

Ranking data were analyzed within each role group. As rankings represent ordinal, within-subject measurements, Friedman tests were used to assess whether participants systematically differentiated among roles within the same group. When significant omnibus effects were observed, mean rank values were examined to interpret relative preference ordering among roles.

In addition to quantitative measures, qualitative data collected through open-ended questions were analyzed to extract representative quotes, providing a richer understanding of participants' opinions.

4 Results

Participants perceived the roles differently across all measured dimensions, with statistically significant effects observed for comfort ($F = 29.14$, $p < 0.001$), emotional support ($F = 18.72$, $p < 0.001$), trust in prediction ($F = 22.76$, $p < 0.001$), advice-following ($F = 22.79$, $p < 0.001$), and desirability ($F = 24.80$, $p < 0.001$). These findings indicate that variations in Likert ratings were systematic rather than random, allowing meaningful comparisons of the distinct perceptual profiles associated with each role.

Descriptive analysis (Table 2) shows that the Visionary, Tester, and Forecaster roles consistently received high ratings in perceived comfort, trust in prediction, advice-following, and desirability. The visionary emerged as the most balanced role, ranking in the top three across all five dimensions. In contrast, the Shadow was the least preferred, ranking in the bottom three across every metric. Similarly, the Twin and Persona were not perceived favorably, particularly regarding desirability and trust.

Open-ended responses help contextualize these evaluations. Several participants described the Shadow as "a little creepy", portraying it as "on your shoulder, always looking at what you are doing", noting that it "didn't offer as much help and guidance" than more proactive roles. The Twin elicited similar discomfort, with concerns about an "exact copy" that "talks and acts like me", described as "eerie" or "too close". The Persona was also viewed skeptically, with some characterized its "ideal" version as a "fantasy horror scenario" that might "eventually take over" or "discourage" rather than motivate the user.

4.1 Perceived Differences among Role Groups

LMM analyses revealed that both the Prediction Maker and Data Explorer group received significantly higher ratings than the Self group for comfort, trust in prediction, advice-following, and desirability. No significant differences were found between the Prediction Maker and Data Explorer groups. Bonferroni-adjusted pairwise comparisons confirmed that these group-level differences were statistically reliable, suggesting that participants' perception aligned with intentional conceptual distinctions between role groups rather than with isolated role features. At the group-level, there were no significant differences in perceived emotional support ($F = 0.365$, $p = 0.694$). To examine whether this null effect masked within-group variability, we conducted a sensitivity analysis excluding the Shadow role, which had the lowest descriptive ratings. After exclusion, the Self group scored significantly higher in emotional support than both the Prediction Maker and Data Explorer groups ($F = 3.509$, $p = 0.03$), while the significance patterns for all other dimensions remained consistent.

4.2 Roles Ranking Priorities

The Friedman tests of the ranking data revealed clear, systematic preference structures within each role group.

For the roles of the Self group, the participants significantly differentiated among the four roles ($\chi^2(3) = 36.05, p < .001$). Self Sculptor emerged as the most preferred role, while Shadow was consistently ranked lowest. Qualitative insights show that Self Sculptor offered the "chance to manipulate the data" and provided "more autonomy over the resulting choices" compared to roles that felt "too prescriptive". One participant stated "it reminds me that I don't have to fix everything at once" and that "making small, consistent changes each day can really make a difference over time".

For roles within the Prediction Maker group, participants also showed strong and consistent preferences ($\chi^2(3) = 63.35, p < .001$). Forecaster and Visionary were most frequently ranked at the top, whereas Future Teller was overwhelmingly ranked last. This pattern might suggest that participants favor predictive roles framed around strategic insight and guidance, over those as fixed or inevitable outcomes. As reflected in the open-ended responses, Forecaster was appreciated for offering "very clear and data-driven insights" that appeal to users who value "facts and numbers", and Visionary was perceived as "inspiring" and "supportive," motivating users to "keep going" by showing what is possible through "consistent

Table 2: Descriptive Analysis Results (Asterisks* and underlines highlight the top 3 and bottom 3 performers in each dimension.)

Roles	Comfort		Emotional Support		Trust in Prediction		Advice-following		Desirability	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Self Sculptor	5.37	1.304	4.66*	1.581	4.97	1.240	5.03	1.249	5.12	1.398
Persona	4.79	1.630	4.29	1.665	4.65	1.438	4.65	1.492	4.58	1.690
Shadow	4.55	1.664	3.86	1.707	4.45	1.429	4.44	1.484	4.30	1.743
Twin	4.70	1.722	4.39	1.628	4.53	1.477	4.56	1.529	4.49	1.769
Visionary	5.46*	1.298	4.56*	1.522	5.05*	1.216	5.16*	1.174	5.25*	1.372
Path Finder	5.35	1.358	4.43	1.575	4.72	1.306	4.86	1.345	5.03	1.471
Future Teller	4.89	1.616	4.03	1.592	4.74	1.463	4.69	1.431	4.66	1.663
Forecaster	5.50*	1.244	4.04	1.548	5.38*	1.189	5.37*	1.206	5.29*	1.364
Librarian	5.32	1.268	4.23	1.504	4.97	1.276	4.96	1.300	5.06	1.338
Living Memory	4.89	1.607	4.61*	1.614	4.64	1.457	4.69	1.423	4.70	1.611
Time Machine	5.11	1.538	4.12	1.615	4.92	1.354	4.89	1.386	4.90	1.594
Tester	5.51*	1.197	4.13	1.588	5.19*	1.269	5.22*	1.267	5.27*	1.369

effort." In contrast, the Future Teller was frequently criticized for being "vague and ambiguous," leaving users with "choice paralysis" and "without really helping [them] what to do right now".

The preferences for the roles within the Data Explorer group were more evenly distributed but still statistically significant ($\chi^2(3) = 15.93, p = .001$). Tester was most frequently prioritized. Participants valued the "low-risk, thoughtful approach" of Tester in the open-ended responses, which allowed them to "experiment with different health strategies safely" without having to "jump in blindly". In contrast, Living Memory tended to be ranked lower despite receiving high Likert ratings for emotional support, which may suggest that emotionally meaningful framings are appreciated conceptually but not prioritized when users must negotiate trade-offs across perceptual dimensions. This is reflected in some participants' responses, characterizing it as "a bit passive" and a "trip down memory lane". While they acknowledged it is "educative in as far as retrospective experiences are concerned", these participants preferred "quicker tools that could help me act" and "to consider some trends in the future".

5 Discussion and Next Steps

Regarding RQ1, our findings indicate that users did not perceive HDT roles as interchangeable; rather, they made systematic distinctions between individual roles and role groups. Within the Self role group, almost all roles, except for Self Sculptor, received consistently lower perceptual ratings. It suggests that roles emphasizing mimicry and strong identity proximity to the person (e.g., Twin and Persona), while conceptually central to the definition of Digital Twin as "an extension of self" [5] or "future data selves" [7], were not necessarily perceived as beneficial by users. Participants appeared to favor roles that provide interpretive distance and support exploratory engagement with health data.

The comparatively higher perceptual ratings of Self Sculptor further suggest that users may prefer HDTs that enable them to "work on themselves" or "take care of themselves" [25], rather than merely "know themselves". Winter et al. [25] have argued similarly, suggesting personal health technologies should support a fluid and multiple conception of the self. In this view, preferred roles seem to emphasize experimentation, multi-path exploration, and future-oriented decision-making, rather than simply replicating existing health states or offering a single health suggestion or outcome. These findings create a tension with prevailing HDT taxonomies and technically driven approaches that prioritize "twinning" and representational fidelity as the primary proposition [2, 24].

At the group level, the lack of statistically significant differences between Prediction Maker and Data Explorer roles further suggests that users may not differentiate HDT roles in line with our three groups. Rather than distinguishing primarily between future-oriented prediction and retrospective data exploration, participants may have perceived both as forms of analytical support that externalizes complexity and provides interpretive assistance. This pattern implies that current HDT groupings may not fully align with the categorization logic of participants. Additional qualitative analysis is needed to understand which interaction dimensions may better capture how users differentiate among HDT roles.

Regarding RQ2, compared to comfort, trust in prediction, advice-following, and desirability, emotional support may not be a primary driver of role preference in preventive health contexts. This pattern reflects that preventive health technologies are often understood as tools for monitoring, forecasting, and sense-making, where perceived reliability of health insights remains important to users. However, it also sits in tension with prior work that emphasizes the role of emotional support in sustaining long-term engagement with health technologies [20]. This tension suggests that future HDT designs should carefully balance affective interaction with analytical credibility, and further research is needed to explore how emotional support can be integrated without increasing perceived identity intrusion or undermining trust or reliability. Furthermore, since HDTs are conceptually defined to draw on highly personal health data ranging from biometrics to behavioral data, they are likely to be approached by questions of boundary, privacy, and user control. When an HDT is framed primarily in affective terms without clearly articulating why data are collected and how they will be used, users may perceive discomfort, distrust, or skepticism on this technology.

There are some limitations of this study. First, since participants were recruited via an English-language crowdsourcing platform,

our sample results in a predominant representation of European and North American perspectives. This limits the generalizability of our findings, particularly for roles like the Future Teller, which may carry distinct cultural connotations that were not fully captured in the current data. Second, while participants demonstrated high digital literacy, their prior knowledge of Digital Twin technology was limited, which may have influenced their baseline expectations and role perceptions. Finally, because the study relied on text-based role scenarios, future studies should involve interactive prototypes or immersive scenarios to explore how interaction design influences user experience and evaluations of HDT roles. Our future work will extend the initial analysis by conducting a detailed comparative analysis of the 12 roles and applying cluster analysis to identify interaction features associated with different user perceptions, complemented by a thematic analysis of open-ended responses to understand underlying preferences and concerns. We will also examine associations between user background and perceptual ratings, with a particular attention to how prior self-tracking experience shapes role perception.

6 Conclusion

This study explored user perceptions and preferences for 12 roles of Human Digital Twins in preventive health scenarios. Our quantitative findings indicate that roles framed as closer to the person (e.g., Shadow and Twin) tended to receive lower ratings on perceived comfort, trust, emotional support, and desirability, despite reflecting the central "twinning" metaphor of HDTs. Also, the divergence between perceived emotional support ratings and intra-group rankings suggests that affective features may not be the primary driver of people's role preferences. These initial findings call for rigorous role selection for HDTs in preventive health contexts. Rather than assuming that greater closeness to the self or stronger emotional support will necessarily be received positively, our results suggest careful consideration of how different role framings influence users' perceptions.

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